



Explainable artificial intelligence for crypto asset allocation

Golnoosh Babaei ^a, Paolo Giudici ^{b,*}, Emanuela Raffinetti ^b

^a PhD in Computer Engineering, University of Pavia, Via San Felice 5, 27100 Pavia, Italy

^b Department of Economics and Management, University of Pavia, Via San Felice 5, 27100 Pavia, Italy

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ABSTRACT

Many investors have been attracted by Crypto assets in the last few years. However, despite the possibility of gaining high returns, investors bear high risks in crypto markets. To help investors and make the markets more reliable, Robot advisory services are rapidly expanding in the field of crypto asset allocation. Robot advisors not only reduce costs but also improve the quality of the service by involving investors and make the market more transparent. However, the reason behind the given solutions is not clear and users face a black-box model that is complex. The aim of this paper is to improve trustworthiness of robot advisors, to facilitate their adoption. For this purpose, we apply Shapley values to the predictions generated by a machine learning model based on the results of a dynamic Markowitz portfolio optimization model and provide explanations for what is behind the selected portfolio weights.

1. Introduction

Cryptocurrency markets are a recent choice for investors. An important reasons for the emerging of this new financial market is the collapse of the banking sector in 2008 and the following low interest rate environment. After the launch of the first cryptocurrency, the Bitcoin, several other cryptoassets have been introduced by improving the existing technology (Chokor and Alfieri, 2021).

Recently, many scholars and investors have been attracted by cryptocurrencies. From a research point of view, the application of machine learning (ML) models (Akyildirim et al., 2021; Alessandretti et al., 2018), network models (Giudici and Polinesi, 2021; Giudici et al., 2021), and portfolio optimization models (Jiang and Liang, 2017) for cryptocurrencies are new established topics.

In particular, Derbentsev et al. (2020) utilized some ML algorithms such as support vector machines, logistic regression, artificial neural networks, and random forests to analyze the predictability of cryptocurrencies at different frequencies, such as daily and minute levels. Comparing the results and performance of these algorithms they concluded that support vector machines can achieve the most accurate results compared to the logistic regression, artificial neural networks and random forest classification algorithms. Another related paper is Valencia et al. (2019) who used available social media data to predict the price movements of cryptocurrencies including Bitcoin, Ethereum, Ripple and Bitcoin using machine learning models. After comparing neural networks, support vector machines and random forest, they showed that Twitter data could indeed be used for cryptocurrency prediction by means of neural network modeling.

In terms of crypto asset allocation management, the Markowitz mean–variance framework for cryptocurrency portfolio was utilized recently by Kurosaki and Kim (2021) and Brauneis and Mestel (2019). The latter authors used data on cryptocurrency prices from 1/1/2015 to 12/31/2017 and compared the risk and return of different mean–variance portfolios for cryptocurrency investments against two benchmarks: a naively diversified portfolio and the CRIX. Their Empirical results showed that portfolios

* Corresponding author.

E-mail addresses: golnoosh.babaei01@universitadipavia.it (G. Babaei), paolo.giudici@unipv.it (P. Giudici), emanuela.raffinetti@unipv.it (E. Raffinetti).

containing several cryptocurrencies have the lowest level of risk and perform better than single cryptocurrencies in terms of Sharpe ratio and certainty equivalent returns. Similarly, [Jiang and Liang \(2017\)](#) applied deep learning and convolutional neural network (CNN), to cryptocurrency exchange data for a portfolio management problem. The proposed model was reinforcement learning in which the maximization of the accumulated return is a reward function of the network. Their proposed approach was compared with 3 benchmarks and 3 other portfolio management algorithms, and led to the conclusion that deep reinforcement was the best performing algorithm.

Despite the high accuracy of machine learning models in portfolio optimization, they are not explainable, as the reasons of their decisions are not clear (see e.g. [Giudici and Raffinetti, 2021](#)). They are like black boxes, that can predict accurately but cannot explain why a decision is made ([Bussman et al., 2021](#)). For example, why the proposed optimal portfolio has a large weight on a certain asset? and why another one is highly diversified among a certain set of assets?

The previous lack of explainability has prevented a wider acceptance and diffusion of machine learning models for portfolio optimizations and, more generally, in finance, particularly because financial regulators are not akin to validate this type of models.

Trying to overcome this problem, explainable artificial intelligence (XAI) have been recently proposed.

While some preliminary study apply XAI to cryptocurrencies, none of them have focused on the application of explainable AI for portfolio management (for more information see [Valencia et al., 2019](#) and [Ahelegbey et al., 2021](#)). Therefore, in this study, we fill this gap and propose an explainable portfolio management approach for cryptocurrencies that aims to strike a balance between predictive accuracy and explainability.

The rest of the paper is organized as follows. In Section 2, the methodological background is explained. Section 3 presents the data employed in the paper. In Section 4, we discuss and present our empirical results. Section 5 concludes by summarizing the findings and giving some future research directions.

2. Methodology

In this section, we explain the theoretical concepts and methods utilized in this paper.

2.1. Portfolio allocation

Portfolio asset allocation is a popular topic in finance that indicates the investment amount which should be shared among different available investment opportunities to diversify the risk of a portfolio and simultaneously receive an acceptable return.

The mean-variance model was proposed by the pioneer of modern portfolio theory, ([Markowitz, 1952](#)), that quantitatively express portfolio selection problem by treating it as a quadratic optimization problem. This model assumes that future asset prices can be accurately reflected by market historical prices, which are defined by a vector of expected returns and a covariance matrix based on mean of return, standard deviation of return, and correlation with other assets. An investor can achieve good investment performance by selecting the right combination of assets to invest in. Tracing the efficient frontier, a continuous curve of mean return and risk intersections that indicates the best investment strategy, allows for higher returns with the same risk rate. In essence, this model seeks a trade-off between risk and return ([Kalayci et al., 2017](#)).

Financial technology (FinTech) innovations are driving the banking sector, with more changes expected in the next few years than in the previous two centuries. One of the most significant changes is the use of advanced technologies, such as machine learning algorithms, to facilitate security trading and advisory services for investors, which are referred to collectively as robo-advisors ([Leow et al., 2021](#)).

Most robo-advisors perform and daily update asset allocation on the basis of Markowitz's model, based on a mean-variance analysis, or a variant of it. However, the same robo-advisors rarely disclose information on how they choose their asset class investment universe or how they estimate variances and correlations between asset classes. They even more rarely disclose their expected return and risk parameters explicitly ([Phoon and Koh, 2017](#)).

In this paper, we aim to fill this gap. Before doing so we briefly revise Markowitz model, using a sample portfolio made of crypto assets to optimize daily crypto asset allocation for minimum volatility.

For a given portfolio p , the return is calculated as follows:

$$E[R_p] = \sum_{i=1}^n w_i E[R_i], \quad (1)$$

Where w_i is a weight in $[0, 1]$ and $E[R_i]$ represents the daily return of a set of cryptoassets, for $i = 1, \dots, n$.

To calculate the risk of a portfolio, the following equation can be used:

$$\sigma_p^2 = \sum_{i=1}^n w_i^2 \sigma_i^2 + \sum_{i=1}^{n-1} \sum_{j=i+1}^n w_i w_j \text{cov}(R_i, R_j), \quad (2)$$

Where $\text{cov}(R_i, R_j)$ indicates the covariance between the return of the i th and the j th cryptoasset.

We assume that a robo-advisory updates daily the proposed portfolio, calculating the portfolio weights on the basis of previous information. We also assume, without loss of generality, that a relevant past time window corresponds to a month.

Let X_i ($i = 1, 2, \dots, n$) indicate the price of an asset, in our case a crypto, on the i th day of the available time horizon. We will use the information of the 30 days before the p th day to calculate the portfolio weights for the p th day. Consequently, if $p = 1$ indicates the first day of available data, the first portfolio is generated at day $p=31$.

For a given day p , with $p > 30$, let $E[R_i]$ indicate the mean daily returns of each crypto asset during the last 30 days. Eq. (1) for the first portfolio would then become:

$$E[R_{p=1}] = w_1 E[R_1] + w_2 E[R_2] + \dots + w_{30} E[R_{30}], \quad (3)$$

Whereas, in terms of portfolio risk, Eq. (2) becomes:

$$\sigma_{p=1}^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + \dots + w_n^2 \sigma_n^2 + w_1 w_2 \text{cov}(R_1, R_2) + \dots + w_{n-1} w_n \text{cov}(R_{n-1}, R_n), \quad (4)$$

Where σ_1^2 is the variance of the first crypto asset, calculated using the previous 30th day. $\text{cov}(R_i, R_j)$ refers to the covariance between the i th and j th cryptocurrency within the same thirty days. Hence, $\text{cov}(R_1, R_2)$ shows the covariance between the first and second cryptocurrencies during the mentioned time period.

Once operational, on day 31st, the robot advisor updates its portfolio weights every day. For example, it calculates a second optimal portfolio on the 32th using Eqs. (3) and (4) with data from day 2 to day 31.

Therefore, we assume that for any given day, after the 31st day in the available data, the robot advisor recalculates the optimal portfolio weights, using the previous 30 days.

A question that naturally arises for the user of a robot advisory platform is which crypto assets are most influential in the determination of the portfolio weights. From this viewpoint, Markowitz' model can be seen as a "black box" which produces an optimal set of weights, without explaining which are the most important determinants of such result. Explainable AI methods can be used for this purpose, to explain the most important factors that affect portfolio weights.

To achieve this goal, we first provide Z scores of cryptocurrencies in each portfolio, Z_i , to provide a measure that considers both volatility and return of these assets as follows:

$$Z_i = \frac{R_i - E(R_i)}{\sigma_i}, \quad (5)$$

where R_i represents the return of cryptos in the p th day. $E(R_i)$ and σ_i are the average return of the corresponding crypto and the standard deviation of the asset return during the previous 30 days of the portfolio generation date.

Furthermore, to have a general measure that presents the overall performance of each portfolio, for each daily portfolio, the Z score, that combines the return and the risk, is calculated as:

$$Z_p = \frac{R_p - E(R_p)}{\sigma_p}, \quad (6)$$

Where $E(R_p)$ indicates the mean of the portfolio returns, and σ_p the standard deviation of the portfolio returns, in the previous 30 days.

2.2. Explainable artificial intelligence

Several efforts have been made in recent years to create intelligent agents capable of explaining their decisions (Townsend et al., 2019). According to Adadi and Berrada (2018), interpretability is concerned with the understanding of a given model at two different granular levels: global (analyzing the whole model behavior) and local (focusing on a specific prediction) (Molnar, 2020). In addition, interpretability techniques can be divided into two types based on their application: model agnostic, which are independent of the model, and specific, which are designed for specific problems. In many applications, machine learning models that are complex black box methods are utilized and provide accurate predictions but these outputs need explanations (for more details see Guidotti et al., 2018).

In this study we utilize a model agnostic XAI approach, the SHapley Additive exPlanations (SHAP) model, introduced by Lundberg and Lee (2017) that is based on the game theory. In SHAP approach, to understand the contribution of each featured explanatory variable, different feature coalitions are considered and the difference in each prediction made by adding a selected feature is its "local" shapley value (Shapley, 1953). The local explanations can be combined in an overall global Shapley value leading to the general contribution of each variable.

Specifically, the effect of each predictor X_k , for $k = 1, \dots, K$, is calculated as follows:

$$\phi(f(\hat{X}_i)) = \sum_{X' \subseteq C(X) \setminus X_k} \frac{|X'|!(K - |X'| - 1)!}{K!} [\hat{f}(X' \cup X_k)_i - \hat{f}(X')_i] \quad (7)$$

Here K represents the number of predictors, X' is a subset of $|X'|$ predictors, $\hat{f}(X' \cup X_k)_i$ and $\hat{f}(X')_i$ are the predictions of the i th observation obtained with all possible subset configurations, respectively including variable X_k and not including variable X_k .

Once Shapley values are calculated, the "global" contribution of each predictor can be calculated averaging the "local" contributions over all predicted values.

There are no reasons to assume that Markowitz asset allocation is a linear function of the returns and, therefore, we assume that the relationship between Z_p and the Z_i of the single crypto assets is non linear.

To estimate such a non linear relationship we can employ a machine learning models, for example a Random Forest model.

In such a model, we consider the daily returns of each crypto asset as the explanatory variables. On the other hand, the Z score of each portfolio is utilized as the response to be predicted.

We will apply Shapley values to the daily series of returns and risks generated by Eqs. (3) and (4).

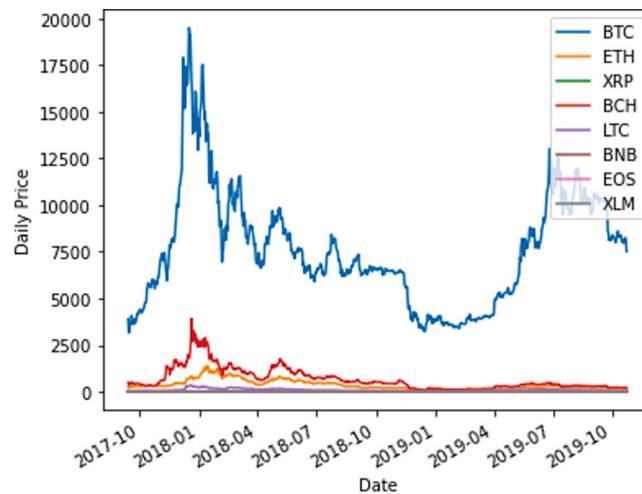


Fig. 1. Price distribution of cryptocurrencies.

Table 1
Descriptive statistics of the selected 8 cryptocurrencies.

Cryptocurrency	Mean	STD	Min	Max
BTC	7597.92	3048.73	3154.95	19497.4
ETH	359.19	258.32	84.31	1396.42
XRP	0.49	0.40	0.16	3.38
BCH	667.82	597.61	77.37	3923.07
LTC	93.92	59.28	23.46	358.34
BNB	13.63	8.81	0.67	38.82
EOS	5.78	3.70	0.49	21.54
XLM	0.18	0.13	0.01	0.89

3. Data and descriptive analysis

To illustrate our approach, we consider the daily price of 8 cryptocurrencies: Bitcoin (BTC), Ethereum (ETH), Ripple (XRP), Bitcoin Cash (BCH), Litecoin (LTC), Binance Coin (BNB), Eos (EOS) and Stellar (XLM) over the period, September 2017 to October 2019 (771 daily observations). Fig. 1 shows how the price of these cryptocurrencies is distributed over the time horizon.

Fig. 1 confirms that BTC has the highest price over the whole period and fluctuates between about 2500 and 20000.

In addition, descriptive statistics of the observations are presented in Table 1.

From Table 1, it can be claimed that BTC, ETH, BCH and LTC are the most volatile cryptocurrencies during the considered time period. Also, the lowest price is attributed to XLM whereas BTC shows the highest price.

4. Empirical finding

In this section, we explain our empirical study and present the results we obtained from applying the proposed model to the selected data.

As explained in Section 2, the main goal of this paper is to propose an explainable portfolio management approach that has the ability to explain the weights assigned by a robot advisor that daily applied Markowitz' asset allocation model to available cryptocurrencies.

For this purpose, we generate portfolios on a daily basis using the historical data of the last 30 days. Therefore, the first day for which we create a portfolio in our data set is 14-10-2017, using the daily price of the eight cryptocurrencies between 14-09-2017 and 13-10-2017.

We then repeat portfolio construction for the 739 consequent available days, thereby generating a total of 740 portfolios.

The portfolio weights can be stored in a matrix which includes 740 rows and 8 columns, representing the sets of weights for each crypto in each built daily portfolio.

We recall that the covariance matrix is updated daily, for each portfolio, using the information of the 30 days prior to the date of the portfolio creation.

Once the portfolio weights are calculated, applying Eqs. (1) and (2) to the observed data, we can obtain the actual return and risk of the 740 portfolios constructed by the robot advisor. To present the characteristics of the generated portfolios by our model,

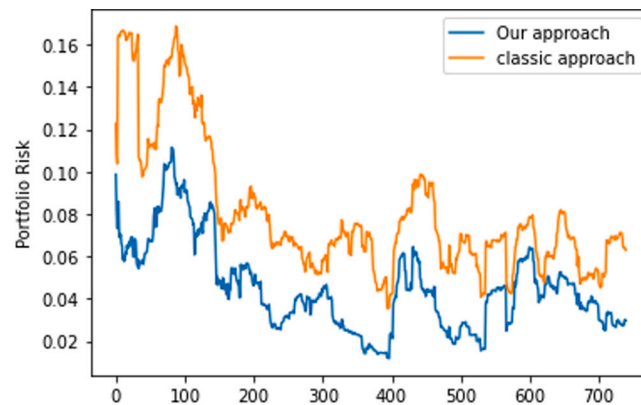


Fig. 2. Comparison of distribution of portfolio risk in our approach and for an equally weighted portfolio.

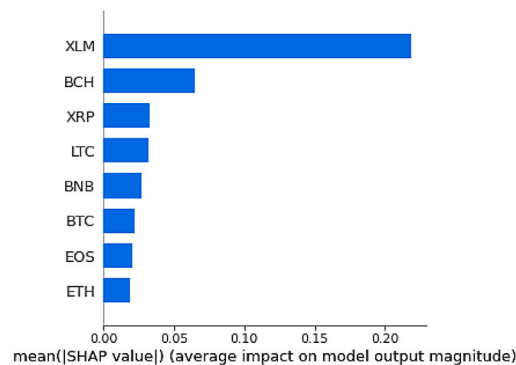


Fig. 3. Overall contribution of cryptocurrencies to the Z scores of portfolios.

Fig. 2 provides the distribution of portfolio risk for the 740 robot advised portfolios, in comparison with the portfolios generated using constant and equal weights.

Fig. 2, shows that the robot advisor allocation strategy, with weights that change over time, leads to portfolios with lower risk levels with respect to a static portfolio, with constant equal weights.

As we explained in Section 2.1, to consider both aspects of a portfolio, we focus on Z score of the portfolios and included cryptos. Therefore, after calculation of risk and return for each portfolio, we find the Z scores using Eqs. (5) and (6).

In the next step, to develop a machine learning model, Z scores of the cryptocurrencies are used as explanatory variables to predict the Z score of the portfolio. For this purpose, to build a random forest model we split the data into train and test using the train ratio of 70 percent. It should be noted that as the focus of this paper is not the prediction accuracy or selecting the best ML model to predict Z score of portfolios, we do not discuss about the model goodness of fit.

Finally, Shapley values corresponding to cryptocurrencies are computed to explain the prediction of our complex ML model. Specifically, using shapley values, we can understand which cryptocurrencies in a portfolio most contribute to explain the Z-score of each portfolio, which is a function of both its return and risk. Fig. 3 presents the overall contributions of each cryptocurrency Z-score to the portfolio Z-score shapley values, averaged across different samples (portfolios).

From Fig. 3 is obvious that Stellar (XLM), followed by, Bitcoin Cash (BCH) and Ripple (XRP) in comparison with other cryptocurrencies are the most explainable variables, that is, those that explain the largest part of the variations of the portfolio Z-scores (functions of both the return and the risk). From an asset allocation viewpoint, this means that the variations in the weights of the optimal portfolio over time are mainly explained by variation in the returns and/or in the risk of these three assets.

Fig. 4 provides a better insight to understand the contribution of cryptocurrencies to portfolio Z score. It provides, for each crypto, and each time point, a color that indicates the magnitude of the corresponding z-score (blue for low values and red for high values) and a location on a horizontal line that indicates the Shapley value of each crypto with respect to the portfolio Z-score.

The plot in Fig. 4 allows a detailed analysis of the local (pointwise) explanation of each crypto, differently from Fig. 3 which presents a global (averaged) explanation. The x-axis represents the shapley values of the 8 cryptocurrencies available on the y-axis. Colorful points on the left show those which shift the response (portfolio risk) in a negative direction while points on the right are the observations that contribute to the predicted values in a positive way.

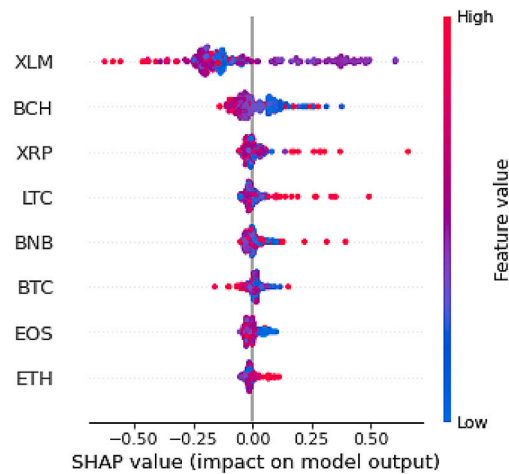


Fig. 4. Contribution of cryptocurrencies to portfolio risk.

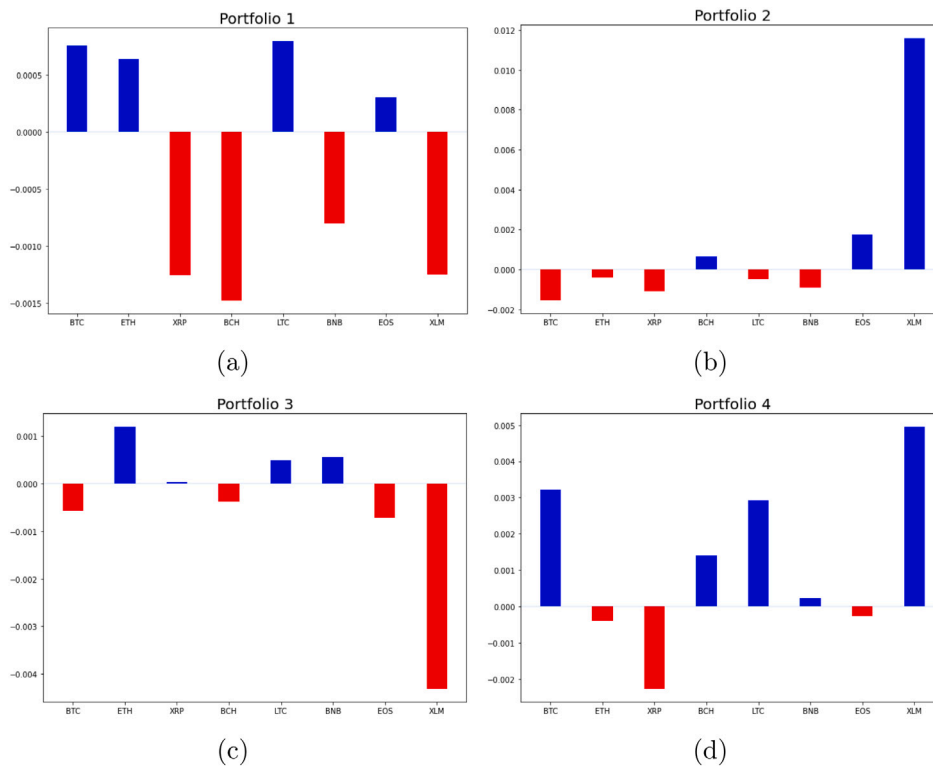


Fig. 5. Contribution of each explanatory variable to the Shapley's decomposition of four generated portfolios. Red color and blue color show negative and positive distance from the mean, respectively.

Fig. 4 indicates that the contribution of XLM is highly variable, leading to high weights for some days and low weights for others. In contrast, less volatile assets such as BTC and ETH provide rather stable weights.

To better explain the “local” explainability of portfolios in Fig. 5 we present the Shapley values of the eight crypto in four portfolios, randomly selected among the 740 proposed allocations.

Fig. 5 clearly shows the advantage of our explainable model. It can indicate which variables contribute more to the prediction of each portfolio's Z-score, based on both returns and risks. Fig. 5 clearly shows how the explanations are different (“personalised”) for each of the four considered portfolios. For example, Portfolio (b) assigns a high value to XLM whereas Portfolio (c) a low value. Generally, it can be said that XLM is a crypto which has the highest importance score in almost all four portfolios, as already discussed.

5. Conclusions

We have presented a methodology that can explain the “automatic” choices of a robot advisor which operates according to Markowitz’ optimal asset allocation.

The methodology has been applied to a time series of portfolios calculated on a set of crypto assets.

The obtained results suggest to implement the method as a useful tool for supervisors, which assess the compliance of robot advisories to financial regulations. But also for robot advisory FinTech companies, which can self-assess the same compliance. It may also be made disposable to investors, through appropriate application service interfaces.

Further research involves the application of the proposed methodology to different portfolio optimization methods, more complex than Markowitz’s, such as those that consider systemic risk and/or tail risk.

Also, other explainable AI methods, such as Shapley-Lorenz decompositions (Giudici and Raffinetti, 2020a), can be applied to find the most explainable determinant of portfolio returns in terms of their predictive accuracy, rather than in terms of their pointwise predictions.

Finally, the proposed methods should also be applied to measure the operational risks that can heavily impact crypto markets and, in particular, cyber risks (see e.g. Giudici and Raffinetti, 2022)

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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